ChatGPT

Implementing Anki-Style Intraday Learning Steps with FSRS Scheduling

Retention Target and Review Frequency in the Current System

Your system currently uses a **target retention of 90%** (meaning it aims for about a 90% recall probability on review) as the default. This is in line with common spaced repetition settings – FSRS itself often uses 0.9 as a default desired retention rate. With this target, intervals are chosen to keep recall high (90%), which can lead to relatively frequent reviews if needed. The **daily review limit has been removed**, allowing unlimited reviews per day (as per the `0014_fsrs6_unlimited_review.sql` migration). This means the system will not forcibly cap the number of reviews – if many cards are due (or if the target retention is set high relative to current performance), users might see a large review load. Be mindful that a high target retention can spike the number of reviews; for example, one user found that raising their target from 83% to 90% caused a surge in daily reviews until the schedule adjusted.

New card introductions are **capped at 10 per day by default**, which is a sensible limit to prevent overload. (The Anki manual similarly warns that adding too many new cards can overwhelm you with reviews later.) Keeping new cards to 10/day ensures that, even with high retention and no review cap, the growth in daily reviews remains manageable as the FSRS algorithm schedules repetitions for those new cards.

FSRS Scheduling vs. Anki's Learning Steps

Currently, your implementation **does not mimic Anki's multi-step learning phase** for new cards. **FSRS (Forgetting Spaced Repetition Schedule) v6** is being used with its default parameters (no personalized optimization yet). On a card's **first review**, you immediately apply FSRS's calculated interval. With default FSRS-6 settings, this yields approximately:

- **Hard → ~1.3 days**,
- **Good → ~2.3 days**,
- **Easy → ~8 days**,

for the next review interval (these are the typical initial intervals given the default stability values). In practice, this means after a new card's first successful review, the user won't see that card again for at least the next day or longer. There is **no same-day repetition** for a new card if the answer was Hard, Good, or Easy – the card is essentially "graduated" into the review schedule immediately with a multi-day interval. Only if the user fails the card ("Again") or triggers a lapse (e.g. opening the card's definition, which you treat as a lapse) will there be a same-day review, and even then FSRS's default lapse interval for a first review is on the order of a few hours (around 5 hours by your notes). In summary, under the current setup a new card is seen **once** on the day it's introduced (unless failed), in contrast to Anki where it's usually seen **multiple times** on day one.

Anki's default behavior for new cards involves **short initial learning steps**. For example, with the default settings, a new card might have *1 minute* and *10 minutes* as learning steps. The user reviews the card shortly after first seeing it: one minute later if they pressed Again initially, then ten minutes later after a successful first recall. Only after these steps does the card "graduate" to a longer interval (commonly 1 day or a few days). This means Anki shows new information twice (or more) on the first day to reinforce it. By comparison, your FSRS-based approach shows new information once (if the user remembers it) and then waits a couple of days. It's not uncommon for Anki's first **graduating interval** to be around 1 day (for Good) and the first **Easy interval** to be ~4 days – but that comes *after* at least one or two same-day reviews (1m, 10m). Users accustomed to Anki might find it jarring that a new word, if marked "Good" in your app, isn't reviewed again until ~2–3 days later, or ~8 days later if marked "Easy". They might worry this is too long without an earlier repetition (indeed, some Anki users express concern when their cards graduate to a week-long interval after only one day's learning steps).

The Case for an Intraday Learning Step ("Match Anki's Feel")

To provide that immediate reinforcement and confidence in remembering, you're considering adding an **intraday (same-day) learning step** for new cards. The goal is to *"match Anki's feel"* by showing new cards more than once on day one. This could help users better consolidate the memory and hit the 90% retention target more comfortably before stretching intervals out.

From a learning perspective, having a short second exposure on the same day can significantly improve initial retention of the material. The FSRS community recognizes this as well: they **recommend adding a single short learning step (10–30 minutes)** when using FSRS, rather than jumping straight to multi-day intervals. By completing all learning steps on the same day, you allow FSRS to start its scheduling with a stronger baseline. In fact, *too many* or too long initial steps can be counterproductive – FSRS can't optimally schedule if the card stays in "learning" for days. The guidance is to keep any learning/relearning steps **within the same day**. Common advice is *not* to use steps longer than ~12 hours, because then some cards wouldn't be seen until the next day, defeating the purpose. A **single 10–30 minute step** is often cited as ideal, as it gives one extra repetition soon after learning, and then FSRS can take over with its calculated intervals.

Using multiple short steps (e.g. Anki's old default of 1m, then 10m, then maybe another) isn't usually necessary with FSRS. In fact, the use of many micro-steps like 5m, 10m, 15m, 30m in a row is **discouraged** by the FSRS developers. The reason is that FSRS's algorithm is designed to determine optimal intervals on its own; if you hold a card in a cycle of many brief reviews, you're essentially doing extra reviews that the algorithm might not need, and it can even lead to anomalies (for example, scenarios where the "Hard" next interval becomes longer than the "Good" interval because the card's been over-reviewed in the short term). However, **one** or at most two intraday steps are fine. If you notice that even with one short step your **young card retention** is below the 90% target, some users add a second same-day step (e.g. a 2-hour or 4-hour review later in the day) to reinforce learning. This is an optional tweak – the majority of users find a single 10–15 minute review is enough for most content, given FSRS will then adapt the subsequent interval based on that success.

In summary, adding an intraday review:

- Gives the learner **early reinforcement** of new material (increasing confidence and retention).
- Aligns with Anki's proven approach for new cards (multiple exposures on Day 1).

- Follows FSRS best practices, which indicate **short same-day reviews** make the scheduling more optimal once FSRS takes over.
- Avoids the feeling of *"I won't see this card again for days, I might forget it"* that some users have when new cards immediately jump to long intervals.

FSRS Algorithm Behavior for Same-Day Reviews

One important aspect is how FSRS itself deals with very short intervals. By default, FSRS's model is primarily built on **daily intervals**; its optimization process used lots of user data where intervals are measured in days. That's why, with default parameters, it proposes intervals of 1+ days for any successful review – it doesn't naturally give a 10-minute interval for a "Good" because it's aiming for ~90% retention and has no concept of a 10-minute test in its core model. However, **FSRS v5 and v6 do have a provision for same-day reviews** built into the algorithm, via a concept sometimes referred to as *"short-term S"*. This is essentially a heuristic adjustment for memory stability (S) when a card is reviewed again on the **same calendar day** [1] .

To explain: FSRS tracks a card's **stability (S)** – a measure of how long the card is expected to be retained – and **difficulty (D)**. After each review, it updates these values. When the next review happens much later (days or weeks), FSRS uses its main formula to update stability. But if a review happens very soon (e.g. 10 minutes or a few hours later, on the same day), FSRS uses a modified formula for that update [1] . The modification ensures a few things:

- Same-day reviews still **increase stability**, but not as much as a next-day review would (because reviewing after 10 minutes doesn't prove long-term memory as much as reviewing after 1+ days). FSRS introduces extra weights (w17, w18, w19 in the algorithm) to scale the stability change for short intervals.
- A successful review on the same day will **not decrease stability**. In fact, FSRS guarantees that if the button pressed is Good or Easy, the new stability S' will be at least as high as it was before that review. (Normally, in the main formula, even Hard is treated as a "pass" that shouldn't reduce stability – and with same-day reviews, they allow Good/Easy to only boost or keep S the same, whereas Again/Hard in the same-day context can lower S slightly if the card proved difficult). This acknowledges that if you remembered the card twice in one day, your memory shouldn't get worse – at worst, it stays the same if the interval was too short to add much benefit.

In practice, what this means is that **when you implement a same-day learning step, FSRS's scheduling formula can accommodate it**. The first review (on a new card) will set an initial stability (S0) based on the outcome. If that first review is followed by a second review on the same day (within hours), FSRS will apply its short-term update formula to adjust the stability. The result is typically that the stability goes up a little bit, but not dramatically, because the interval was very short [2] . For example, if a card was somewhat difficult (low S) and you review it again quickly, w19 in the formula makes sure the increase in S is larger for smaller current S – so **quick repetitions help most for low-stability (new) cards**. By contrast, if a card already had high stability and you review it again the same day, you wouldn't gain much (diminishing returns), which intuitively makes sense.

Key point: Up to now, your system likely hasn't leveraged this aspect of FSRS, because you haven't been doing same-day reviews except on failures. By introducing an intraday step, you'll be operating in this territory. FSRS's math will handle it behind the scenes as long as we feed it the correct information (i.e. that the next review happened X minutes later on the same day).

Implementation Plan: Adding an Intraday Learning Step in Supabase (Postgres)

To integrate a short initial review step into your current setup, the changes will primarily involve the database functions (`handle_review`, `get_next_word`, etc.) that schedule cards. Here's a breakdown of what needs to be done:

1. Detect new card's first successful review:
In the `handle_review` function (which processes a user's answer and computes the next review time and updated FSRS fields), add logic to identify when a card is in its **initial learning phase**. This could be determined by checking if the card has no prior stability/difficulty set, or a review count of zero, etc. When such a card receives a **passing grade** (Hard, Good, or Easy – anything except Again), flag this scenario for special handling. Essentially, "we just had the first successful review of a new card."

2. Override the FSRS interval with a short delay:
Instead of using FSRS's suggested interval for this first review, **schedule a brief intraday interval**. For example, you might choose a default of **10 minutes** for this step (which is within the recommended 10–30min range). This means the card's next due time will be `now + 10 minutes` (or whatever duration you set). In the database, you'll update the card's due timestamp to this short interval. Mark the card's state as something like "learning" or "intraday pending" – you might use a boolean or a special review status for this. (If your schema doesn't currently distinguish learning vs review cards, you may simulate it by, say, keeping stability null or using a separate `phase` column, etc.)

- *Note on the grade handling:* If the user pressed **Easy** on the first review, you might still want to give them the 10-minute review (rather than skipping it). In Anki, pressing Easy during learning can sometimes graduate a card early, but since our goal is a uniform single extra step, you'll likely apply the short step to Hard/Good/Easy all alike (except Again). You could consider making the short interval slightly shorter for Hard (since "Hard" means the user struggled – maybe 5 minutes instead of 10), and slightly longer for Easy (maybe 15–20 minutes since it was easy). But these micro-adjustments may not be necessary; a flat 10 or 15 minutes for any successful first attempt is simple and effective.

- If the user pressed **Again** on a new card (meaning they failed the first attempt), currently FSRS would treat that as a lapse and schedule the card about 5 hours out (which is later the same day). You might want to override this as well for a true first failure. In Anki, if you get a card wrong immediately, it usually just stays in the learning queue – e.g. if the steps are 1m 10m, an initial "Again" shows the card 1 minute later. You could implement something similar: e.g. for a new card failure, schedule it a few minutes later (like 5 minutes) as a retry. This would give the user another chance soon, rather than waiting hours. Since FSRS isn't really tuned for "initial fail → 5 hours" (that 5h is likely calibrated for lapses of cards that were already in memory), a shorter retry makes sense in the learning phase. This can be done by a similar override: if grade = Again on a brand-new card, set due = now + 5 minutes (or even 1 minute). Essentially, treat "Again" as the first learning step in that case.

3. Use the database to track learning vs review status:
Ensure there's a way to know the card is in the middle of its intraday learning. For instance, you might add a

column `learning_step_due` or a boolean `in_learning`. The `get_next_word` function (which retrieves the next card to study) should prioritize cards that are due for a short learning step. In Anki, learning cards due in the next 20 minutes are shown even before reviews; you can adopt a similar approach so that when the 10 minutes are up, the card appears promptly for the user. Since you removed the daily review limit, you don't need to worry about that interfering – just make sure learning cards show up as soon as they're due (and are not counted against new card limits of 10 per day, since they've already been counted when first introduced).

4. Process the second review as the hand-off to FSRS:
When the user does the second review (the one after the 10-minute delay), this is when you **integrate with FSRS as if it were the first real review**. In `handle_review`, you'll detect that a card is currently in the intraday learning state. Now, depending on the outcome of this review, you update the card's FSRS parameters:

- **If the user gets it correct on this second review** (which is likely, since it's a short interval; if they remembered it at 10 minutes, that's a good sign), you will now calculate the next interval using the FSRS algorithm. Use the **grade from this second review** as the input to FSRS's formula for updating stability (S) and difficulty (D). In effect, treat this as the card's "first *graduating* review." For example, if the user presses Good on this review, FSRS will compute a new stability S based on whatever initial S0 was and the 10-minute gap. If you didn't calculate an S0 earlier, you can do it now: FSRS needs an initial stability to start from. There are a couple of ways to handle it:
    - *Option A:* **Calculate initial S0 on the first review but don't use it for scheduling.** You could run the FSRS formula when the card was first reviewed (before the 10-min step) to get an initial stability and difficulty, but then override the interval. This means you have stored S and D after the first review. Then when the second review comes (10 minutes later), you feed those S, D as the "previous" values into FSRS, along with the actual interval (0.007 days) and the grade. FSRS will apply its short-term update formula in this case, yielding a new S. This approach is a bit complex but truest to the FSRS model – it lets FSRS know there was a prior review the same day.
    - *Option B:* **Treat the second review as the first time FSRS sees the card.** This is simpler: don't record any FSRS data on the first review; only when the card comes for the 10-minute review do you initialize FSRS. In this approach, you effectively ignore the first review's grade in terms of FSRS calculations. You would assign a default initial difficulty (D0) and stability (S0) now based on the second review's grade. FSRS's official method to get initial S involves curve-fitting as described in the algorithm write-up, but practically, default parameters give fixed initial S values for each grade (e.g. something around 2.5 days for Good, 1.3 for Hard, 8 for Easy as noted). If the user got it right twice in a row, it's safe to assume the card isn't extremely difficult, so starting FSRS as if this were the first review should be fine. The slight downside is that FSRS doesn't "know" the card was seen 10 minutes earlier, but given that 10 minutes is so short, the effect is minimal – it might slightly overestimate stability, but the algorithm's short-term heuristic would have made only a small adjustment anyway.

Regardless of A or B, once this second review is processed, assign the card its next interval from FSRS. Now you'll see something like: maybe Hard = ~1.3 days, Good = ~2 days, Easy = ~7–8 days (the numbers might come out a bit different if FSRS factored in the short gap). The card is now effectively **graduated from**

learning. You should clear the "in_learning" flag and store the updated stability (S) and difficulty (D) in the database. The card's next due date is set to that interval in the future.

- **If the user fails on the second review (Again on the 10-minute review):** This indicates the card is still not sticking. In this case, you can treat it similarly to an Anki lapse in learning: perhaps send the card back to the first step. For example, you could schedule it again in 5 minutes and not consider it graduated. Essentially, the card stays in the learning phase until it gets a successful review. This might be a rare case, but it's good to handle it: you don't want a card that was seen, failed, seen 10 min later, failed again, to then jump to FSRS's 5-hour lapse – better to keep it cycling on a short loop until the user gets it right, or maybe give up for the day if it keeps failing (that's a design decision – Anki has a limit on how many times a card can loop in learning in one day to prevent frustration).

5. Adjust the scheduling functions accordingly:
The changes above mean your SQL functions need to incorporate a bit more logic:

- `handle_review` should set the next due time differently for new cards vs review cards. It will also need to handle updating (or not updating) FSRS fields depending on the phase.
- `get_next_word` (or whatever function fetches the next due card) should be aware of intraday due cards. Likely, you want it to serve any card whose due timestamp is in the past (or due within the next few minutes) regardless of whether it's a learning card or a normal review. If you want to exactly mimic Anki's behavior, you could ensure learning cards due *very soon* are shown before regular reviews. But since you have no daily cap, the ordering is less critical. Just don't leave learning cards waiting too long, or the benefit of the short step is lost.

Keep in mind that you'll also need to consider the UI side: the front-end might want to display the remaining learning step or show that the card is in learning mode (e.g. Anki shows an orange indicator or a count of learning cards). At minimum, ensure that the user can't accidentally bury the learning card – since you removed review limits, as long as the due time is correct, it will appear.

6. Optional – User setting for intraday steps:
As you mentioned, you could make this intraday step configurable. Some users might prefer the current "straight to FSRS" approach, especially if they are learning very easy material or are okay with longer initial intervals. Others (likely most) will prefer having that extra review. A toggle in user settings (e.g. *"Enable short initial learning step"*) could switch this behavior. Implementation-wise, that toggle would just bypass the new logic if turned off (i.e. if off, do exactly what you do now; if on, apply the above overrides). Default it to on, since it's generally recommended by both user experience and FSRS guidelines.

7. Choose sensible defaults for the interval:
The exact duration of the intraday step can be decided based on intuition and FSRS community feedback. The **FSRS Helper guide suggests 10–30 minutes** as a reasonable single step. Many users choose **15 minutes** as a happy medium. This is short enough to reinforce the card while the memory is fresh, but long enough that the user has likely seen a few other cards in between (preventing simple rote memorization). It also ensures that even if someone is studying right before bed, a 15-minute step won't push the card past midnight (which could complicate the "same day" condition). You could go with **15m** as default. If you allow customization, let advanced users pick from say 5m, 10m, 15m, 30m, or so. Avoid anything beyond an hour as a "learning step", because then some users might, for example, study in the morning and then a 4h step would be due in the afternoon – that's fine, but if someone studies in the evening, a 4h step might land

after they've stopped for the day and cross into the next day (which is not a true intraday step anymore). So, consistency matters here.

8. Lapse/relearning steps:
Your question mainly focuses on new cards, but a quick note: if you later consider lapses (relearning), a similar principle can apply. FSRS by default handles lapses via the algorithm (reducing stability, etc.). Anki, however, has relearning steps (e.g. if a review card is forgotten, it might go through a 10m step again). The FSRS forums also recommend keeping relearning steps within the same day. For now, your implementation already treats a lapse like a 5h schedule. You might eventually decide to shorten that for consistency (maybe a 10m refresh then back to review). This would be a parallel change to consider for the future, using the same mechanism you build for new cards.

Ensuring FSRS Parameters Still Work Well

You mentioned that **no personalized parameter fitting** is implemented – you're using the stock FSRS-6 parameters (the `fsrs6Defaults` in your code and the `fsrs6_parameters()` in the database). These defaults were learned from a large dataset (hundreds of millions of reviews) and are a decent generic fit. When you add an intraday step, you are not fundamentally changing those parameters, just the sequence of reviews. In fact, the default parameters themselves assume users might be doing short initial steps (since many Anki users in the dataset would have 10-minute learning steps). The algorithm's short-term review heuristic is already part of FSRS, so the parameters (w17, w18, w19, etc.) related to same-day reviews will kick in automatically. You therefore do **not** need to tweak any FSRS weights immediately.

Over time, once you accumulate enough data, you could run the FSRS optimizer on your users' review logs to get personalized parameters (or even a single adjusted set that better fits your user base). That could further improve scheduling accuracy. But that's independent of the intraday step feature – the intraday step will make the early-stage reviews more effective (preventing some simple lapses), which should actually help the optimizer later by providing clearer success data for new cards. For now, sticking with the default FSRS parameters is fine, and you should expect the algorithm to handle the new cadence without issues.

Final Thoughts and Next Steps

By introducing a short intraday learning step, you'll bring your app's behavior closer to the expected norm in spaced repetition software, providing a gentler onboarding for new information. This addresses the gap where a brand-new word might otherwise be seen once and then not again for several days. With a 10–15 minute second exposure, users get that extra practice while the memory is fresh, which can significantly boost retention (helping meet that 90% target). Meanwhile, FSRS will continue to handle the long-term scheduling, now with the benefit of an extra data point from day one.

From an implementation standpoint, the change requires careful adjustments in the Postgres functions to ensure smooth state transitions (new → learning → review). Be sure to test the following scenarios thoroughly after implementing:

- New card answered Good/Easy on first try → comes back in X minutes → answered again → next due in a few days (graduated).

- New card answered Hard on first try → comes back in X minutes → answered Good (or Hard/Easy) → next due computed correctly (and Hard on first try might be an edge case to test if you decide to vary the short interval for Hard).
- New card answered Again (failed) on first try → comes back in Y minutes (short retry) → perhaps failed again or succeeded, etc. Ensure it doesn't schedule out incorrectly.
- Cards in learning should not count towards the "10 new per day" if they are already introduced (they're essentially the same card just pending). Also, make sure that if a user exits the app and returns after a while, the due learning cards appear as expected.

Overall, this enhancement is expected to **increase initial retention** and user satisfaction, as it combines the strengths of Anki's immediate reinforcement with FSRS's adaptive scheduling. It's a practical step supported by the research and community guidelines, and it should make your application more effective for learners. Good luck with the implementation!

Sources:

- Anki Manual – *Learning Steps and Graduating Intervals*
- Anki Forums – *FSRS Helper: Learning-Steps Recommendations* (advice on using a single 10–30min step with FSRS)
- *A Technical Explanation of FSRS* – Expertium's Blog (details on FSRS's handling of same-day reviews in FSRS-5/6) [1]
- Anki Forums – *FSRS - Reviews drastically up* (discussion of retention setting and review load)
- Reddit - *FSRS default parameters and usage* (note on default parameters sourced from large dataset)

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[1] [2] A technical explanation of FSRS | Expertium's Blog
https://expertium.github.io/Algorithm.html